Session Objectives

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- Distinguish between the therapies used to manage latent vs. active tuberculosis.
- Describe RIPE treatment for tuberculosis. Explain the mechanisms of action, resistance, and significant adverse effects associated with these drugs.
- Review the second-line drugs used for tuberculosis, including aminoglycosides and fluoroquinolones.
- Describe the drugs used to manage nontuberculosis mycobacteria. Review the macrolide class of protein synthesis inhibitors
- Identify the drugs used in leprosy management, including dapsone and the highly teratogenic, thalidomide.

Source: Course Syllabus

Tuberculosis

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- High global mortality rates (underdeveloped areas, drug resistance)
- HIV comorbidity is a challenge
- At least 6 months required for effective treatment (multi-drug therapy)
- Goal: kill bacilli rapidly, prevent relapse, and disease transmission.
- Directly observed therapy (DOT) is necessary.
- Multidrug resistance: resistant to isoniazid or rifampin. Requires increased treatment and use of second-line drugs

Latent Tuberculosis

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- No symptoms
- Skin or blood test indicates TB; Normal chest x-ray and a negative sputum test
- Inactive TB in body, no spread of infection
- Requires drug prophylaxis
- TNF alpha inhibitors (etanercept, adalimumab) activate latent infections (Screen patient prior to initiation).

Active Treatment

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- **Isoniazid (INH), Ethambutol (EMB), Pyrazinamide (PZA), Rifampin (RIF). Treat with all four drugs for 2 months. Continue INH + RIF for 4 months (6 months total)**
- **Pregnancy: INH, RIF, EMB for 2 months. Continue INH + RIF twice weekly for 7 months. Not PZA or Streptomycin.**
- **Pediatrics: RIPE for 2 months, Continue INH + RIF for 4 months (6 months total). Non-severe TB can be treated for 4 months.**

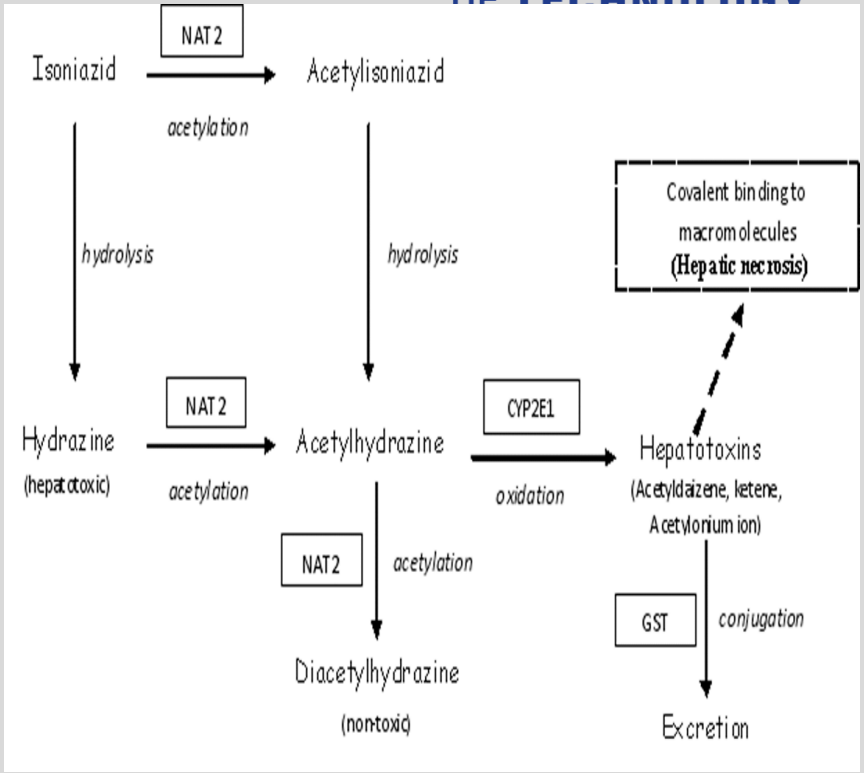
Isoniazid (INH)

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- Well-absorbed
- Structural similarity to pyridoxine
- Bactericidal; Inhibits synthesis of mycolic acid
- Activated by catalase-peroxidase (needs katG)
- Resistance results from mutated katG
- Acetylation (Drug-induced lupus)
- Adverse Effects: hepatotoxicity, especially in alcoholics, plus receiving PZA and RIF
- Peripheral neuritis (manage by B6; pyridoxine administration, 10-50 mg/day).
- Ethionamide similar in MOA (Second-line for TB)—activated by EthA

Isoniazid hepatotoxicity

- Acetylated toxic free radicals have been consistently implicated in the pathogenesis of INH hepatitis
- Drug-induced liver injury secondary to INH is more frequent in alcoholics or those taking rifampin and pyrazinamide.
- Fatal: requires liver transplantation.



Pyrazinamide (PZA)

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- Bacteriostatic, but bactericidal on replicating bacteria
- Prodrug, needs Pzase for pyrazinoic acid conversion.
- Adverse effects: hepatotoxicity, hyperuricemia
- Resistance: mutations in genes involved in drug bioactivation by Pzase (PncA gene)

Rifampin (RIF)

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- Widely distributed into fluids (adverse effects)
- Inhibits DNA-dependent RNA polymerase
- Active against gram +, -, and acid-fast bacteria
- Prophylactic for *N. meningitidis, H influenzae*
- Adverse effects: Red secretions, hepatitis, CYP 3A4 inducer (Revs metabolism)
- Resistance: alteration in the DNA dependent RNA polymerase enzyme.
- "3 RRRs"



Second-Line Drugs and MDR strains

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- Aminoglycosides: capreomycin, amikacin, streptomycin (parenteral)
- Rifabutin
- Rifapentine
- Fluoroquinolones (Moxifloxacin)
- Azithromycin
- Ethionamide
- Cycloserine
- Bedaquiline: MDR strain
- Pretomanid: MDR strain
- Linezolid: MDR strain

  "BPaL regimen"

Aminoglycosides

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- Adverse effects: ototoxicity, nephrotoxicity, and neurotoxicity, teratogenic.
- Resistance: drug unable to bind ribosome, decreased uptake of drug, increased aminoglycoside modifying enzymes (acetyltransferases, adenyltransferases, phosphotransferases).

Aminoglycosides

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- **Treats: Gram-negative aerobic rods**
- **Post-Antibiotic Effect : Inhibits the growth of bacteria after removal of the drug. Immune system has time to remove bacteria.**
- **Concentration-dependent: Increased drug concentration compared to MIC = greater bactericidal action.**
- **Aminoglycosides and beta lactam antibiotics are synergistic and bactericidal (used for *Staph. Aureus*)**

Aminoglycoside adverse effects

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Neuromuscular Blockade

- Decreases acetylcholine release from the pre-synaptic neuron
- Avoid in anesthesia with neuromuscular blockade, myasthenia, and Parkinsonism.

Ototoxicity

- Toxic to both hearing and balance.
- Overlap toxicity with another ototoxic drug (loop diuretics, cisplatin)
- Cranial nerve 8 damage (pregnancy)

Source: Course Syllabus

Aminoglycoside adverse effects

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Nephrotoxicity

- Acute Tubular Necrosis (ATN). Loss of urine concentrating ability. Non-oliguric renal failure.
- Caution in patients taking other nephrotoxic drugs*

Source: Course Syllabus

Rifapentine and Rifabutin

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- **Rifapentine: treats MAC, *M. tuberculosis.* Potent CYP induction and adverse effects as rifampin. Long half-life (allows for once-per-week dosing).**
- **Rifabutin: treats MAC, *M. tuberculosis.* Less CYP induction than rifampin. Replaces rifampin for HIV patients treated with HAART therapy to reduce drug interactions. Similar hepatotoxicity to rifampin. Reduce dose if CrCl is <30 mL/min.**

Fluoroquinolones

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- **Adverse Effects:** Increased QT interval
- Damage growing cartilage
- Category C (pregnancy)
- Tendon rupture
- Photosensitivity
- **Drug interactions**: CYP 1A2, di, trivalent cations
- **Resistance:** Reduced uptake of antibiotic, alteration of DNA gyrase

Moxifloxacin, Levofloxacin

- "Respiratory" fluoroquinolones
- *Legionella, Mycoplasma, Chlamydia, Klebsiella, Mycobacterium avium-intracellulare, M. tuberculosis, H. influenzae, S. pneumoniae.*

Source: Course Syllabus

Linezolid

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- Binds 50S subunit
- Gram + activity
- Use: penicillin resistant *S. pneumoniae* and methicillin and vancomycin-resistant *Staphylococci* and vancomycin-resistant *Enterococci.* Multidrug resistant strains of TB
- Good GI absorption. 100% bioavailability.
- IV and oral dosing on same schedule.
- Adverse effects: Weak MAO inhibitor, caution with serotonin-increasing drugs. Bone marrow suppression.
- Resistance: mutation in 23S rRNA gene

Bedaquiline

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- Inhibits ATP synthase in mycobacteria
- Active against replicating and non replicating bacilli
- Metabolized by CYP P450 system (drug interactions)
- Long $t_{1/2}$ life
- Adverse effects: hepatotoxicity and cardiac toxicity (Increased QT interval-Blackbox warning)

Pretomanid

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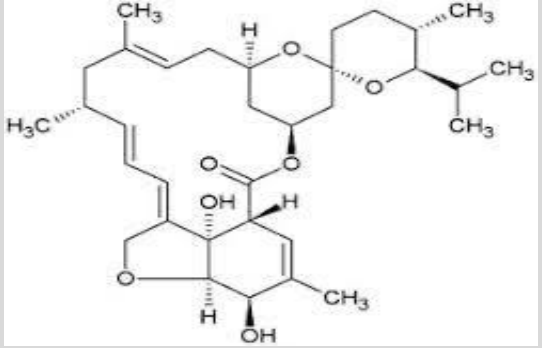
- Nitroimidazole (prodrug), activated by the Ddn enzyme. ”think metronidazole”
- Increases NO to disrupt mycolic cell wall.
- Combined with bedaquiline and linezolid- **BPaL**
- Treat patient for ~ 6 months
- Indicated for pulmonary extensively drug resistant (XDR), or treatment-intolerant or nonresponsive multidrug-resistant (MDR)
- Long $t_{1/2}$
- Adverse effects: neuropathy, increased QT interval, hepatotoxicity

Mycobacterium avium complex (MAC)

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- **Presentation: fever, night sweats, weight loss, increase alkaline phosphate, anemia.**
- **Azithromycin or clarithromycin, plus ethambutol. Rifabutin can be added as a 3rd agent.**
- **Prevents MAC in HIV patients with a $CD_4 < 50$.**

Macrolides (Azithromycin, Clarithromycin, Erythromycin)



- Bacteriostatic Proteins Synthesis Inhibitors
- Macrolide because of chemical structure (Macrocyclic lactone)
- Binds to 50S ribosome; Inhibits translocation
- Eythromycin is a CYP 3A4 inhibitor
- Macrolides safe in pregnant and pediatric patients
- Use: *Chlamydia, Mycoplasma, Legionella, Corynebacterium Diphtheria*, pertussis, syphilis.
- Resistance: efflux of drug, methylation of the 23S ribosomal binding site

Macrolides

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Clarithromycin:

- More active against *Strep.* and *Staph*. *H pylori, Moraxella cattarhalis, Legionella, M. pneumoniae, Borrelia*
- Well absorbed, less GI effects than erythromycin.
- Twice daily dosing
- CYP inhibition

Azithromycin (Zithromax®):

- Increased activity against *H. Flu, Salmonella/Camphylobacter, Chlamydia*
- Long $t_{1/2}$, Once a day dosing
- No CYP inhibition

Hansen disease

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- *Mycobacterium leprae*
- Leprostatic drugs: Dapsone, Rifampin, Clofazimine, or Thalidomide.
- Dapsone used for *Pneumocystis* prophylaxis in HIV patients. Topical use for severe acne.
- Thalidomide for erythema nodosum leprosum (TNF-alpha suppression).

Dapsone

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- **Resistance:** Alteration of dihydropteroate synthase, decreased antibiotic uptake, increased PABA production.
- **Adverse effects: (similar to sulfonamides):** crystalluria, interstitial nephritis, G6PD deficiency, kernicterus (neonate), rash (erythema multiforme), photosensitivity, hypersensitivity, cross allergy with certain diuretics, hypoglycemic drugs

Clofazimine

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- Lipophilic, long duration of bactericidal action
- MOA: anti-inflammatory action (suppresses T-lymphocytes); Binds to guanine of DNA.
- Adverse effects: Discolors skin and conjunctiva, GI upset, increase QT interval. Not in pregnancy.
- Not for commercial use.
- Physician required to register as an investigator with the NHDP. When approved, the medication is dispensed from NHDP to the prescriber (National Hansen's Disease Program)

Thalidomide

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- Reduces the levels of TNFα, decreases angiogenesis. Treatment of multiple myeloma, leprosy
- Fetal effect: Phocomelia (short or absent long bones of the limbs).
- Drug is racemic; (R) enantiomer is effective against morning sickness but the (S) is teratogenic
- 1st trimester danger

